Main energy loss mechanisms for different particles

Photons

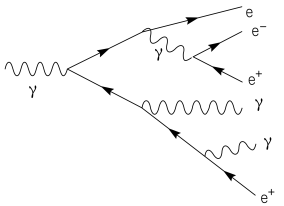
- **Pair production** (dominant at high energy)
- **Compton scattering**
- **Photoelectric effect**

Charged particles

Electrons **Muons** **Protons** **Pions** **Kaons**

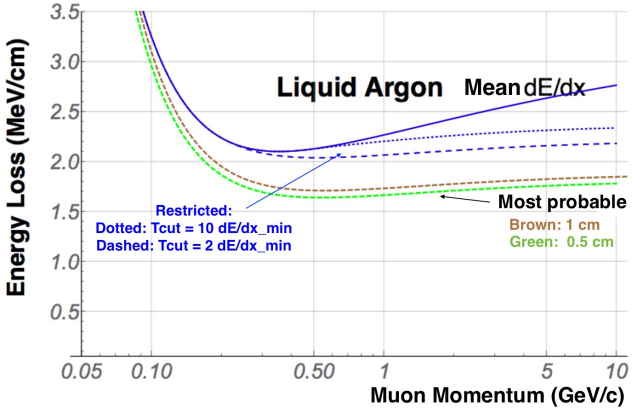
- **Bremsstrahlung** (dominates above critical energy depending on material. Energy loss is proportional to $1/M^2$ → the lighter the particle, the greater the acceleration → important for electrons)
- **Ionization**
- **Protons/pions/kaons can interact with nuclei (→ hadronic showers)**

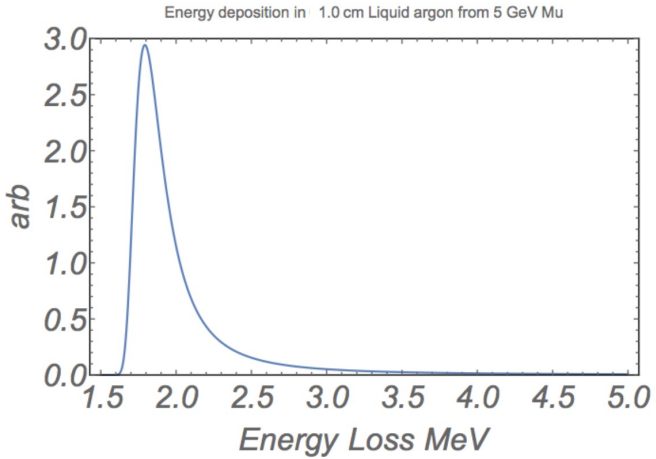
Photon pair production + electron bremsstrahlung = EM shower



Energy loss by ionization

- Charged particles moving through matter ionize the atoms. Mean rate of energy loss per unit distance (*stopping power*) by ionization given by Bethe-Bloch equation.
- At minimum, stopping power ~ independent of particle type and material
- Cosmic ray muons are minimum-ionising particles. The mean energy loss is about 2.1 MeV/cm





Mean energy loss is weighted by very rare events with large single-collision energy deposits. Most probable energy loss is lower than mean energy loss

https://lar.bnl.gov/properties/pass.html#:~:text=Useful%20numbers%20for%20liquid%20argon,(1.5)%20MeV%2Fcm.
Particle Data Group (PDG): https://pdg.lbl.gov/2016/reviews/rpp2016-rev-passage-particles-matter.pdf

Bethe-Bloch equation

$$\left\langle -\frac{dE}{dx} \right\rangle = Kz^2 \frac{Z}{A} \frac{1}{\beta^2} \left[\frac{1}{2} \ln \frac{2m_e c^2 \beta^2 \gamma^2 W_{\max}}{I^2} - \beta^2 - \frac{\delta(\beta\gamma)}{2} \right]$$

Red curve below same as what is in plot on previous slide, but multiplied by density. Units: MeV cm² / g

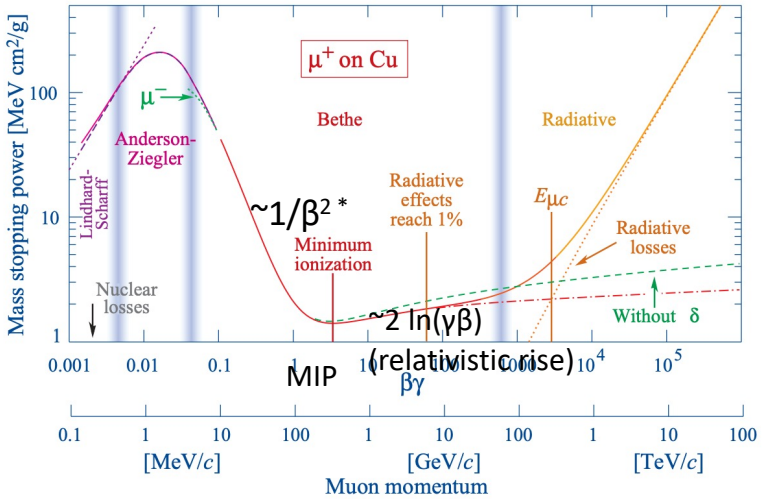
$\beta = \frac{p}{E}, \gamma = \frac{E}{M}$ relativistic factors

$K = 4\pi N_A r_e^2 m_e c^2$

Maximum energy transfer in a single collision for particles of mass M

$$W_{\max} = \frac{2m_e c^2 \beta^2 \gamma^2}{1 + 2\gamma m_e/M + (m_e/M)^2}$$

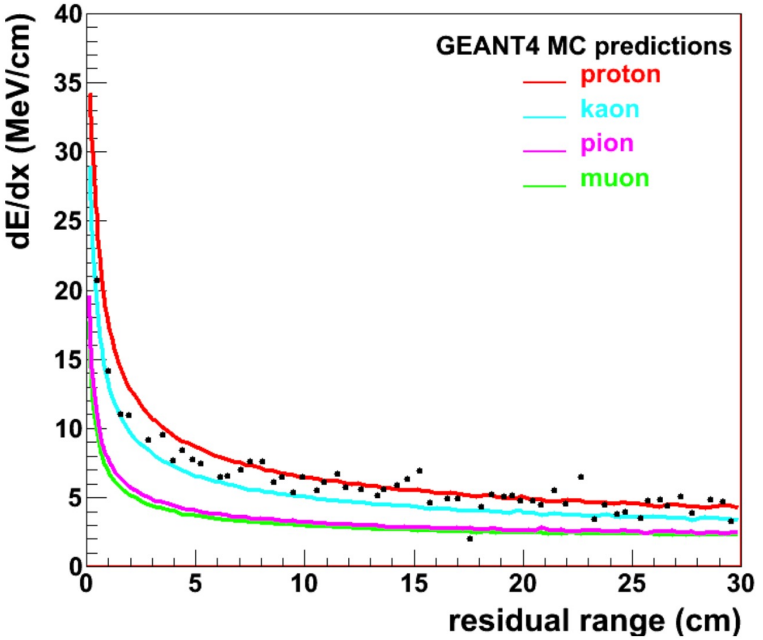
Z = atomic number of absorber (number of protons)
A = atomic mass of absorber (number of protons + neutrons)
z = charge number of incident particle
I = mean excitation energy
$\delta(\beta\gamma)$ = density effect correction due to polarisation

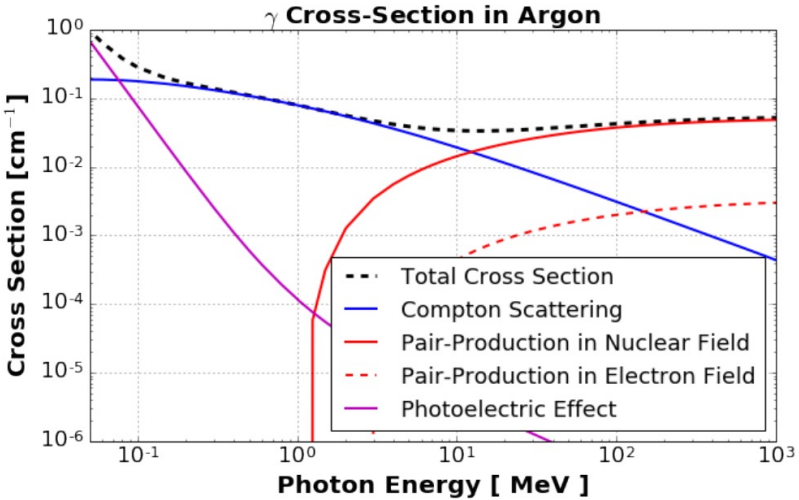


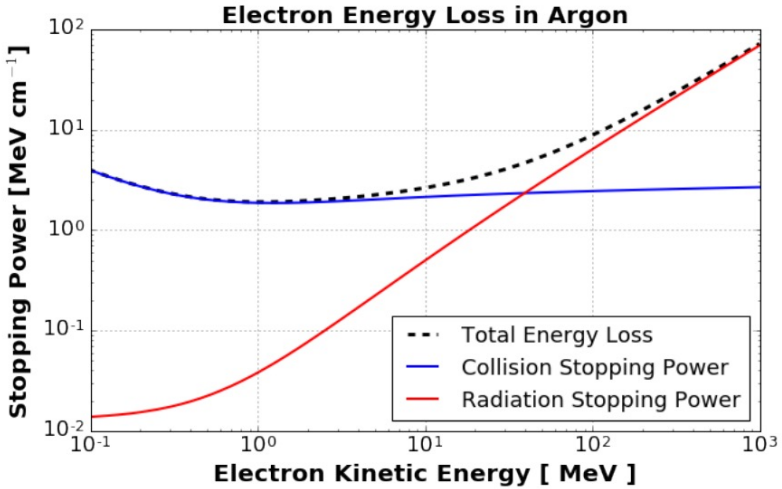
https://lar.bnl.gov/properties/pass.html#:~:text=Useful%20numbers%20for%20liquid%20argon,(1.5)%20MeV%2Fcm.
Particle Data Group (PDG): https://pdg.lbl.gov/2016/reviews/rpp2016-rev-passage-particles-matter.pdf

Particle identification

- Charged particles lose energy by ionisation
- Mean energy loss by ionisation as a function of residual range (remaining length of trajectory) can help discriminate between particle types




γ Cross-Section in Argon
Cross Section [cm⁻¹]
Photon Energy [MeV]
Total Cross Section
Compton Scattering
Pair-Production in Nuclear Field
Pair-Production in Electron Field
Photoelectric Effect


Electron Energy Loss in Argon
Stopping Power [MeV cm⁻¹]
Electron Kinetic Energy [MeV]
Total Energy Loss
Collision Stopping Power
Radiation Stopping Power